LIVE SESSION PLAN

HANDSHAKE-001

A pre-registered single-shot test for information arriving from the future

T0 · THE DRAWS

10:00

Monday 12 October 2026 · German time (CEST) · 08:00 UTC

T1 · THE TARGET

13:00:00

Monday 12 October 2026 · German time (CEST) · 11:00 UTC

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Pre-registration post	x.com/DavonGluck/status/2108645176260313365
Code and data	github.com/hanzunye/handshake-001 · DOI 10.5281/zenodo.23269344
Protocol SHA-256	f4c83cd7d93c9c90e2a3439ebfd833d271f8cde8678ec000b51cdd96db01fb06
Code SHA-256	21352f760902d51c952bfa12c0fccea06ad3434d246e43decaf6d2ae89c38c17

WHAT HAPPENS TODAY

The experiment in one minute

- **10:00** Two quantum random numbers are drawn, 24 bits each (6 hex characters): one from the **ANU** quantum random number service, one from **24 qubits on an IBM Quantum** processor.
- Each number is sealed with a secret 256-bit salt and two hashes (SHA-256 and SHA3-512). **Only the hashes are posted on X.** The numbers stay hidden.
- **13:00:00** The **NIST Randomness Beacon** publishes a new random value that nobody can predict. The target is its first 6 hex characters.
- A draw is a **HIT** only if its 6 characters equal the target exactly. After the target is public, the numbers and salts are revealed and anyone can check them.

1 in 16,777,216
chance of a hit, per draw

≈ 1 in 8,388,608
chance that at least one draw hits

$2^{-48} \approx 3.6 \times 10^{-15}$
chance that both draws hit

Expected outcome: **INCONCLUSIVE** (no hit). A single miss refutes nothing; a hit would be treated first as a possible error.

MONDAY 12 OCTOBER 2026

Today's schedule

CEST	UTC	STEP	COMMAND ON SCREEN
09:25	07:25	Stream starts; pre-registration and file hashes shown	`shasum -a 256 …`
09:45	07:45	Preflight self-check (clock, NIST, ANU, IBM, code)	`python3 handshake.py preflight`
10:00	08:00	**T0:** ANU number drawn and sealed; IBM job sent	`python3 handshake.py start`
~10:01		ANU commitment posted on X	X post
~10:02		IBM number sealed; IBM commitment posted on X	`python3 handshake.py commit-ibm`
10:30–12:50		Waiting; both numbers stay hidden	—
13:00:00	11:00:00	**T1:** the NIST pulse of this second is the target	beacon.nist.gov
13:01	11:01	Target fetched and posted on X	`python3 handshake.py fetch-target`
~13:05		Reveal: numbers, salts and result published	`python3 handshake.py reveal`
~13:10		Check against the hashes on X; end of stream	`python3 handshake.py verify …`

FIXED IN ADVANCE · ENFORCED BY THE CODE

Rules and deadlines

09:30–09:58 07:30–07:58 UTC	Preflight window: a passing self-check must run entirely inside it
09:59–10:10 07:59–08:10 UTC	The only window for `start`, which runs **only once** (ANU: up to 10 attempts)
12:30 10:30 UTC	The IBM job must have **finished** (IBM server time; created by 12:25)
12:50 10:50 UTC	Both commitments must be **posted on X**; a late post voids that draw
13:01 11:01 UTC	Target fetched from this minute on; if NIST is late we wait, it is **never replaced**

> If one draw fails, it is void and the other continues alone. If the attempt cannot start (for example a failed preflight or a power cut), it is postponed by exactly **24 hours**, announced on X before 10:10 (at most 7 times). Every attempt, void and postponement is published.

T1 · 13:00:00 CEST · 11:00:00 UTC

The target

- The target pulse is **the first NIST Randomness Beacon 2.0 pulse stamped at or after 13:00:00 CEST** (11:00:00 UTC) on 12 October 2026.
- The target is the **first 6 characters of its `outputValue`**, in uppercase. NIST is the only source; there is no backup source.
- Anyone can open the same pulse, now or years from now (this page shows "404 not found" until 13:00):

  `https://beacon.nist.gov/beacon/2.0/pulse/time/1791802800000`

Secondary analysis (separate, not live-streamed)

- Three later pulses are compared with the values already made public: **N2 ≈ 14:00 today**, **N3 ≈ Tue 13 Oct 13:00**, **N4 ≈ Mon 19 Oct 13:00** (CEST).
- They ask whether the beacon itself can be influenced. They never change the primary result.

TRANSPARENCY

What you will see, and how to check it

On screen

- Every command and its full output
- The commitment hashes and the X posts with their times
- The NIST pulse on NIST's own website
- At the reveal: both numbers, both salts, the result

Not shown before the reveal

- The two numbers and their salts: never displayed, stored only in private files and backed up offline
- API keys and the IBM job result page are never shown

Check it yourself

- Instructions: `VERIFY.md` in `github.com/hanzunye/handshake-001`
- One command recomputes every hash, fetches the target from NIST and recomputes the result:
  `python3 handshake.py verify --anu-sha256 … --ibm-sha3-512 …`
- Use the hashes **as posted on X**, not the copies in the repository.